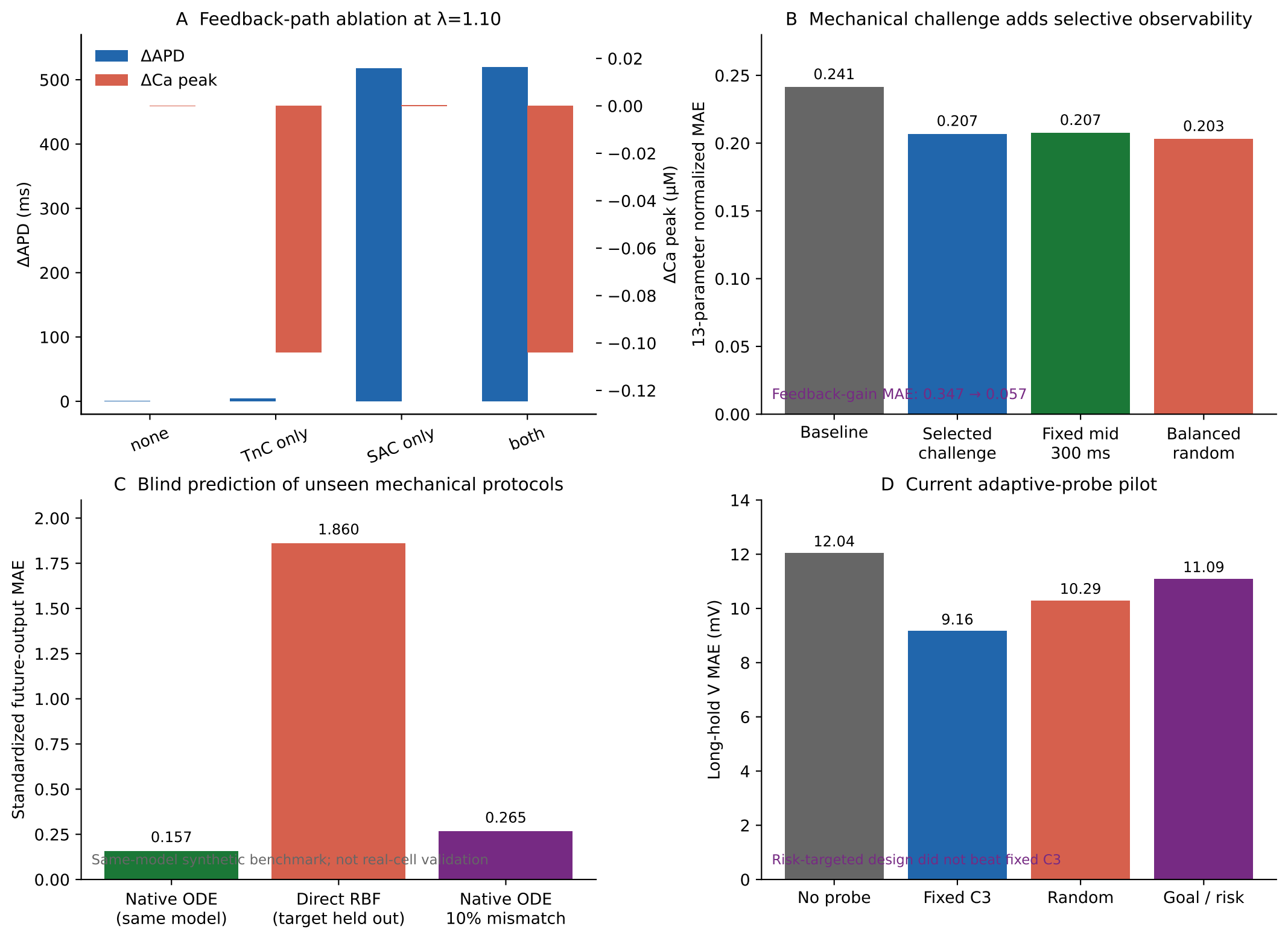
A Feedback-path ablation at λ=1.10
ΔAPD
ΔCa peak
ΔAPD (ms)
ΔCa peak (μM)
none
TnC only
SAC only
both
B Mechanical challenge adds selective observability
13-parameter normalized MAE
0.241
0.207
0.207
0.203
Feedback-gain MAE: 0.347 → 0.057
Baseline
Selected challenge
Fixed mid 300 ms
Balanced random
C Blind prediction of unseen mechanical protocols
Standardized future-output MAE
0.157
1.860
0.265
Same-model synthetic benchmark; not real-cell validation
Native ODE (same model)
Direct RBF (target held out)
Native ODE 10% mismatch
D Current adaptive-probe pilot
Long-hold V MAE (mV)
12.04
9.16
10.29
11.09
Risk-targeted design did not beat fixed C3
No probe
Fixed C3
Random
Goal / risk